The Cost of Intelligence

AIML-501 | Artifact 8 | Explanatory document, classification rationale & reflection

Purpose and central argument

This visual artifact explains the end-to-end development cycle of generative large language models (LLMs) and makes the main resource burdens understandable to a nontechnical audience. Its central argument is that training cost is not a single figure: the full system involves data, compute, energy, time, and recurring evaluation, alignment, and deployment costs.

How the information was classified

I used two distinct classification systems to avoid conflating process with resource accounting. First, the process is organized chronologically: source data → preparation and governance → pretraining → alignment and instruction tuning → evaluation → deployment and monitoring. A feedback loop links downstream failures to upstream data and tuning decisions. Second, resources are divided into four categories: (1) data inputs and curation, (2) computation and supporting infrastructure, (3) energy and environmental overhead, and (4) elapsed time and iteration capacity. This separation makes it clear that a stage can consume several resources simultaneously.

Rationale for the visual design

A left-to-right flowchart communicates sequencing, while four equal-sized resource panels show that the cost drivers are complementary rather than competing stages. The model comparison uses consistent labels and explicit disclosure notes instead of a misleading dollar bar chart. The relationship page emphasizes that scale only creates value when paired with data quality, architecture, training efficiency, safety, and evaluation. The dark navy, teal, and blue palette and compact typography mirror the existing portfolio style; every panel includes descriptive text so color alone never carries meaning.

Model evidence and interpretation

Meta Llama 3.1 405B: Meta documents roughly 15 trillion pretraining tokens, 30.84 million H100 GPU-hours and 8,930 metric tons of location-based CO2-equivalent emissions for this model. Meta also discusses scaling beyond 16,000 H100 GPUs. GPU-hours measure aggregate device utilization, not elapsed training duration. The environmental figure should not be interpreted as direct electricity consumption.

DeepSeek-V3: The technical report states 14.8 trillion pretraining tokens and 2.788 million H800 GPU-hours across the official training stages. The quoted $5.576 million equals 2.788 million × an assumed $2/H800 GPU-hour rental price. This is a compute rental estimate, not the company’s total cost of developing the model; earlier experiments, research, staffing, data work, and serving are outside that boundary.

OpenAI GPT-4: OpenAI describes pretraining on token prediction and subsequent reinforcement learning from human feedback, but its GPT-4 technical report withholds specific model size, training compute and hardware detail. Rather than inserting third-party speculation into a direct comparison, the visual marks

those values as “not disclosed.” These examples illuminate reporting differences; they do not establish a ranking of overall training cost or model quality.

Key considerations and limitations

Compute totals across H100 and H800 processors cannot be compared as equivalent dollar amounts without consistent utilization, accounting, hardware, and time boundaries. Facility electricity, power usage effectiveness, cooling, and location-based greenhouse gas estimates measure related but different phenomena. Development costs also extend beyond the final training run to research, discarded trials, data work, safety testing, labor, and inference. Public reports are incomplete and may use different definitions; therefore, uncertainty and omissions are displayed instead of imputed.

Learning reflection

Building this artifact changed how I interpret claims about the cost of AI. Initially, GPU-hours and a published training price seemed like straightforward ways to compare models. Examining the primary technical reports showed that those metrics often cover different hardware and accounting scopes. The most challenging design decision was how to represent unavailable information without making the graphic feel incomplete. I addressed that by treating disclosure itself as a meaningful comparison category and by separating training stages from resource categories. I also learned that visual clarity can improve analytical rigor: showing feedback loops and resource dependencies makes it harder to overlook iteration, evaluation, and downstream serving costs. This connects directly to applied machine learning and operational strategy, where leaders must distinguish measurable operating inputs from uncertain total investment before making technology decisions.

References

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